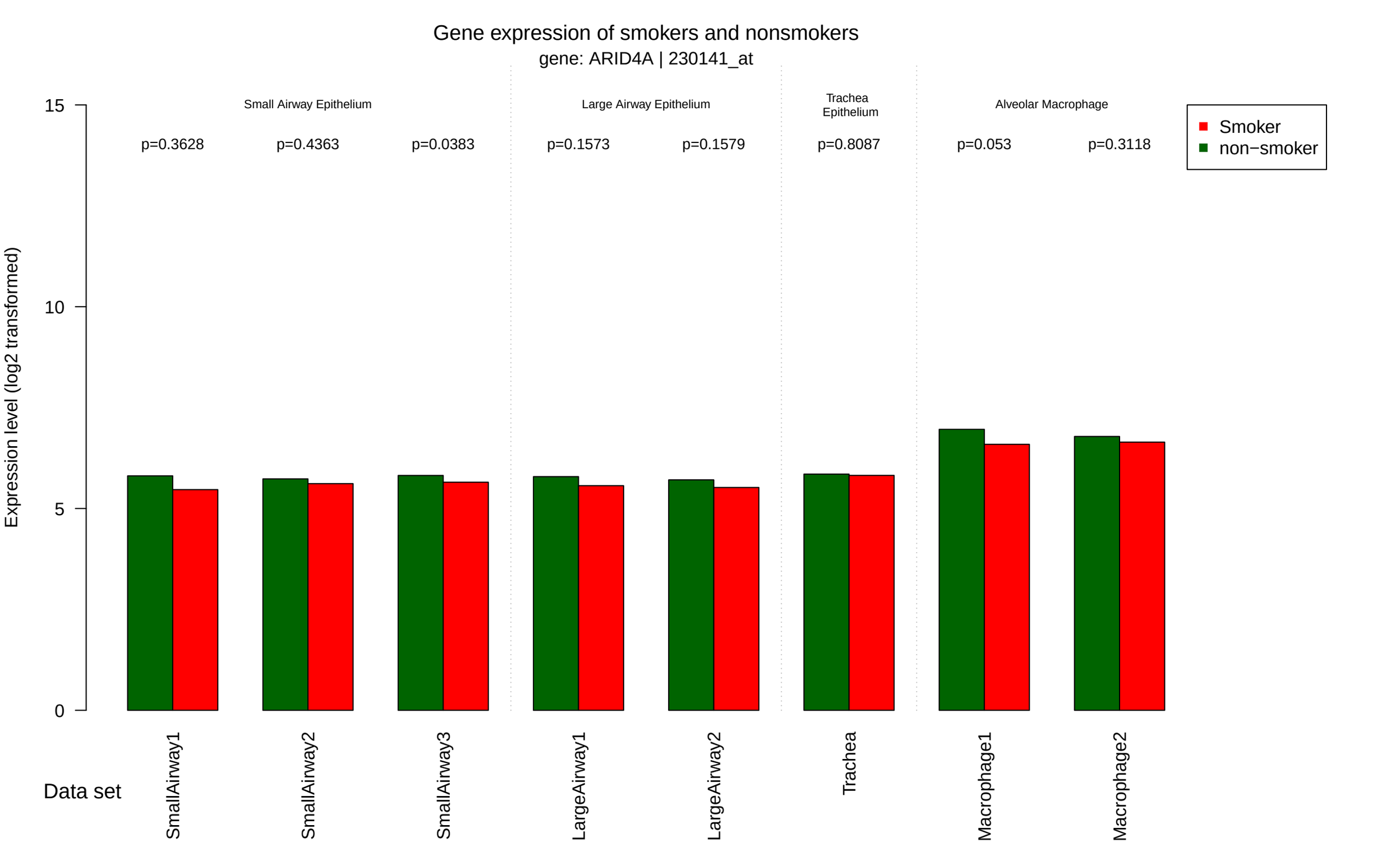
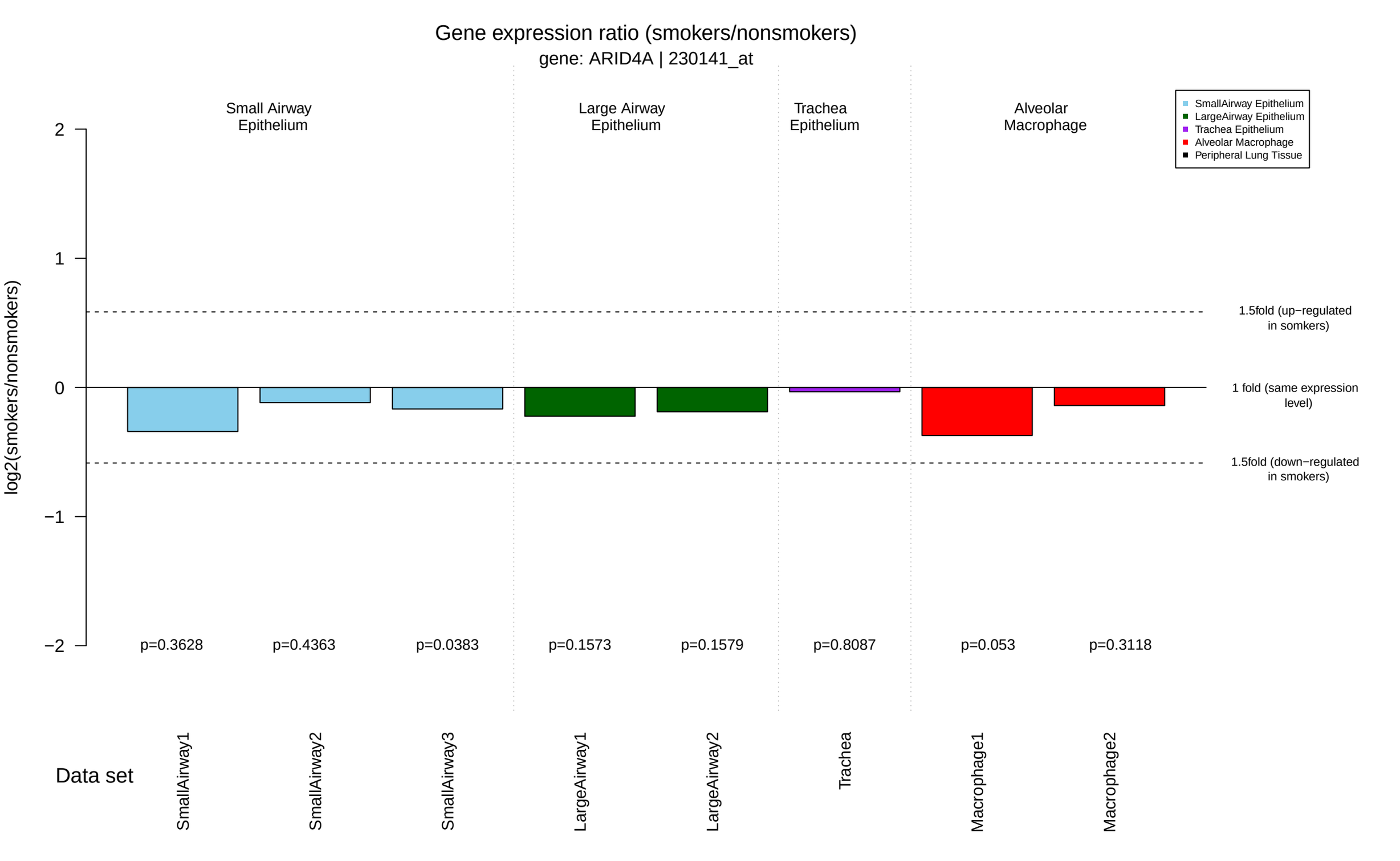
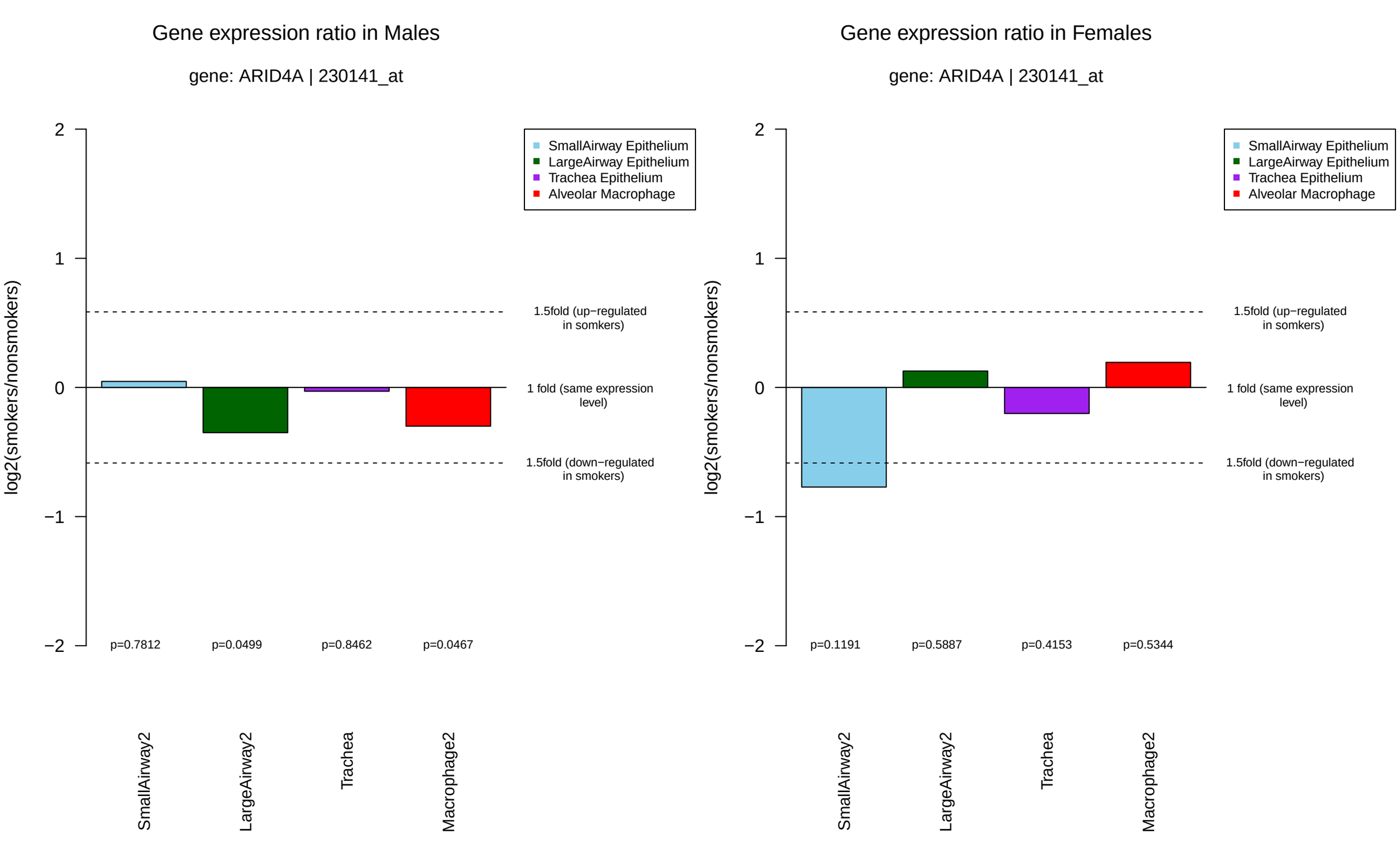
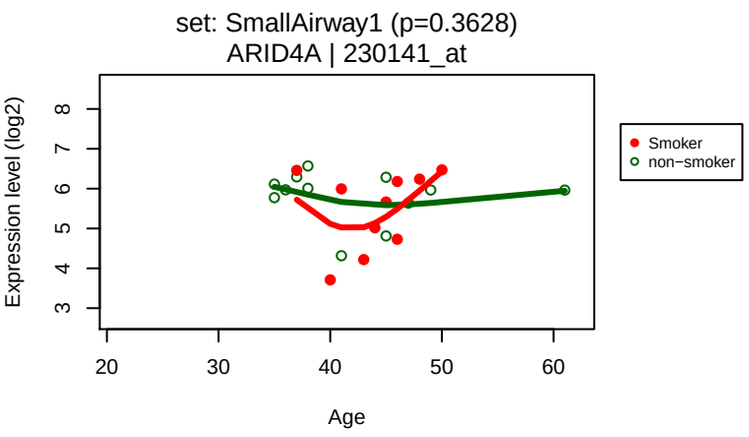
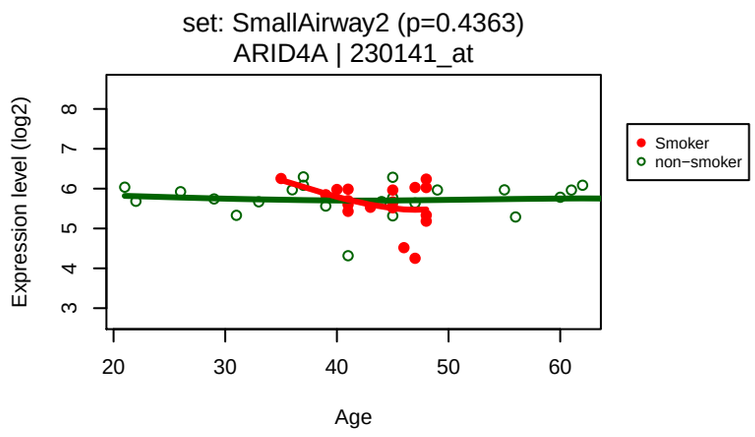
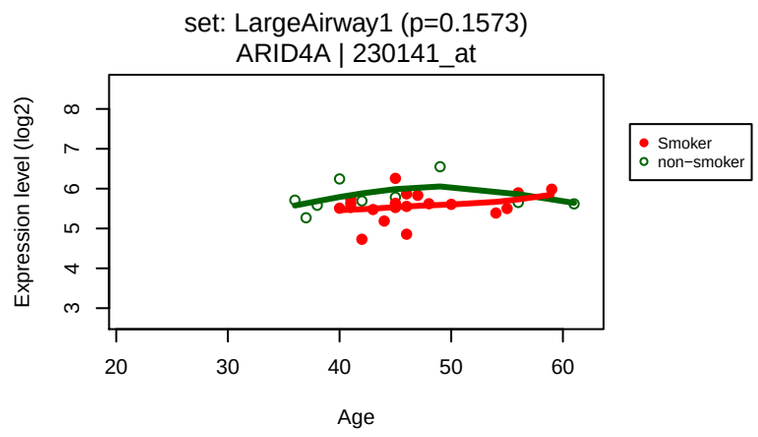

Gene expression of smokers and nonsmokers
gene: ARID4A | 230141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3628
p=0.4363
p=0.0383
p=0.1573
p=0.1579
p=0.8087
p=0.053
p=0.3118
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

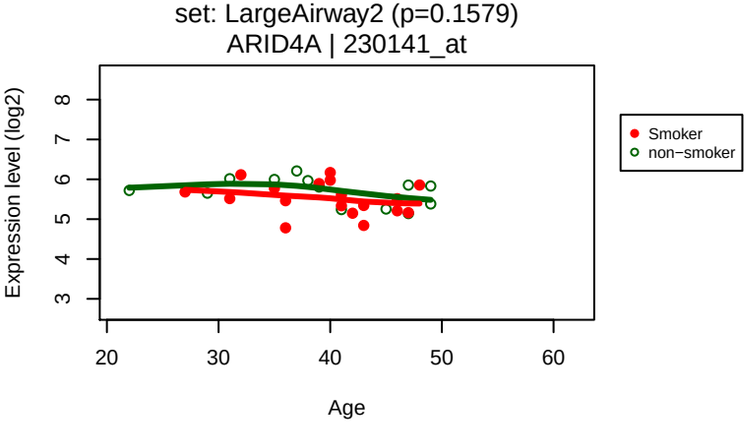

Gene expression ratio (smokers/nonsmokers)
gene: ARID4A | 230141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3628
p=0.4363
p=0.0383
p=0.1573
p=0.1579
p=0.8087
p=0.053
p=0.3118
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

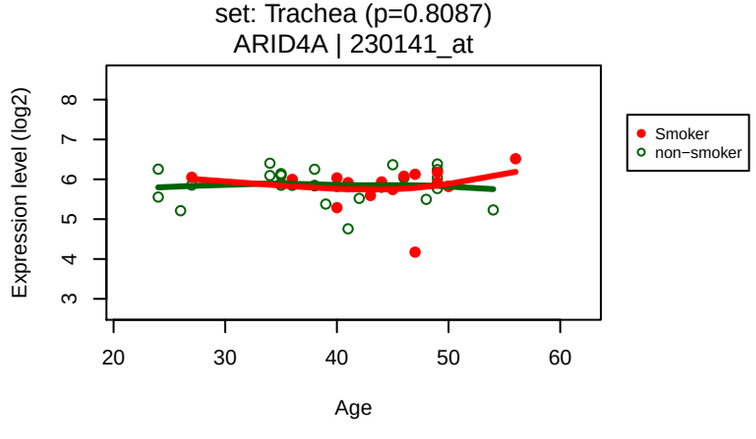

Gene expression ratio in Males
gene: ARID4A | 230141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7812
p=0.0499
p=0.8462
p=0.0467
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARID4A | 230141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1191
p=0.5887
p=0.4153
p=0.5344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

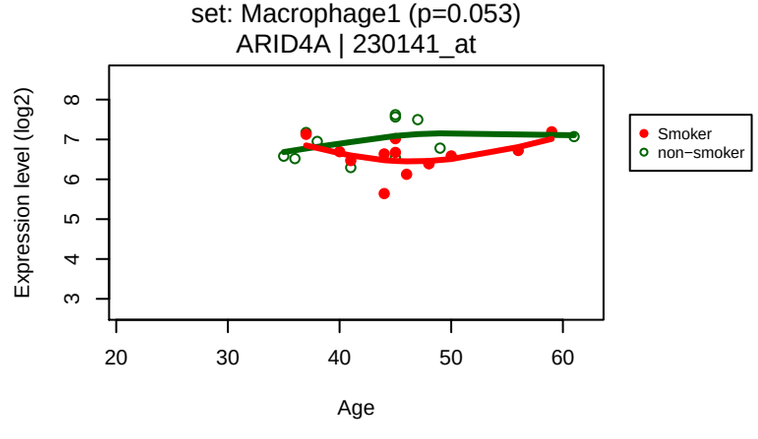

set: SmallAirway1 (p=0.3628)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker

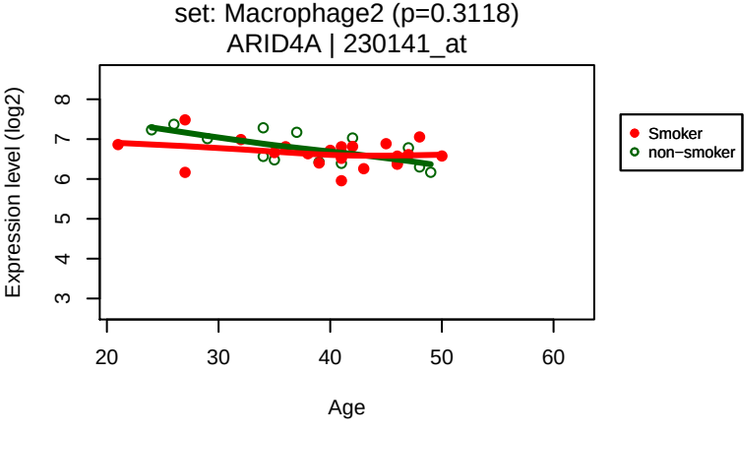

set: SmallAirway2 (p=0.4363)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker

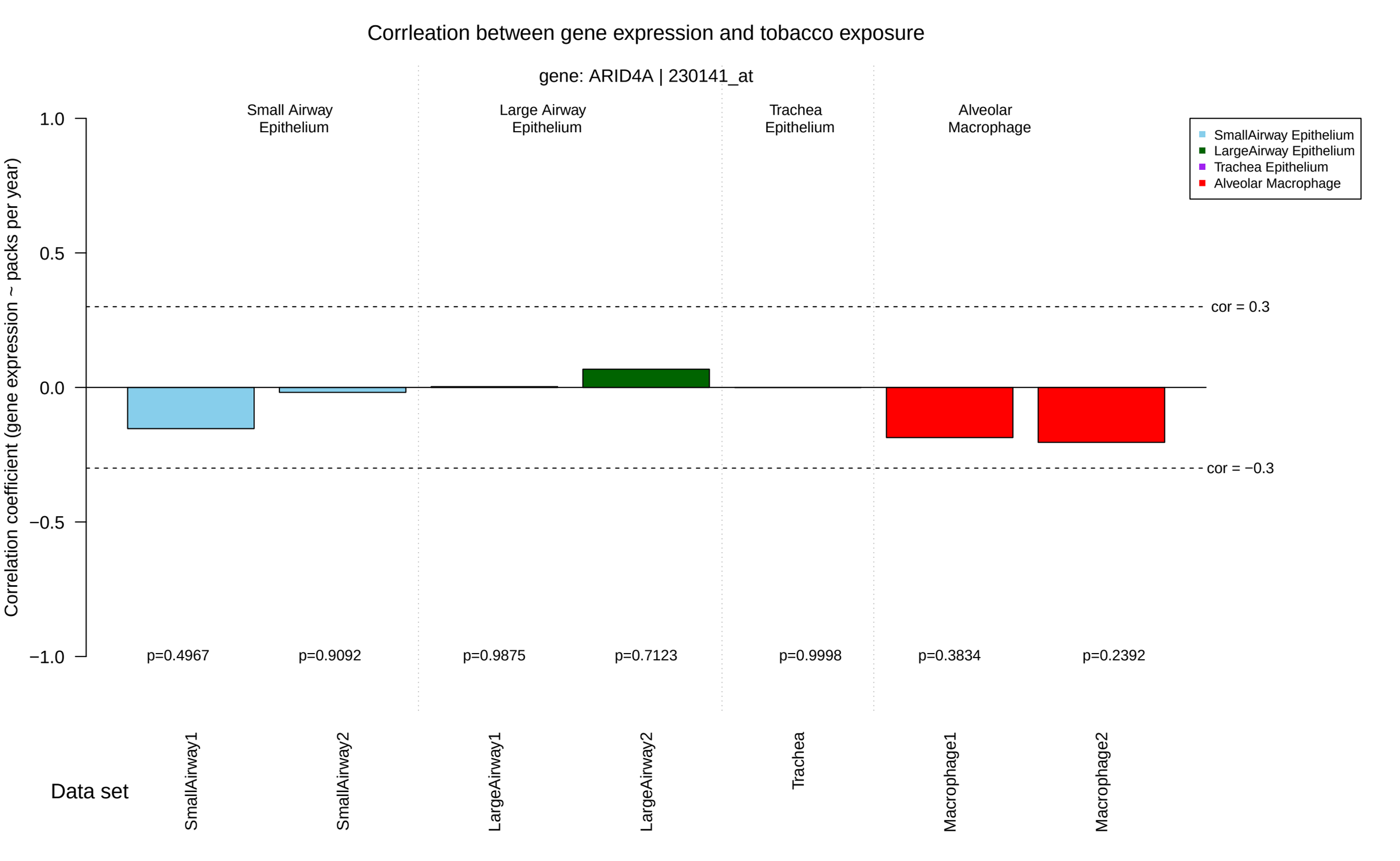

set: LargeAirway1 (p=0.1573)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1579)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8087)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.053)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3118)
ARID4A | 230141_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARID4A | 230141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4967
p=0.9092
p=0.9875
p=0.7123
p=0.9998
p=0.3834
p=0.2392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2